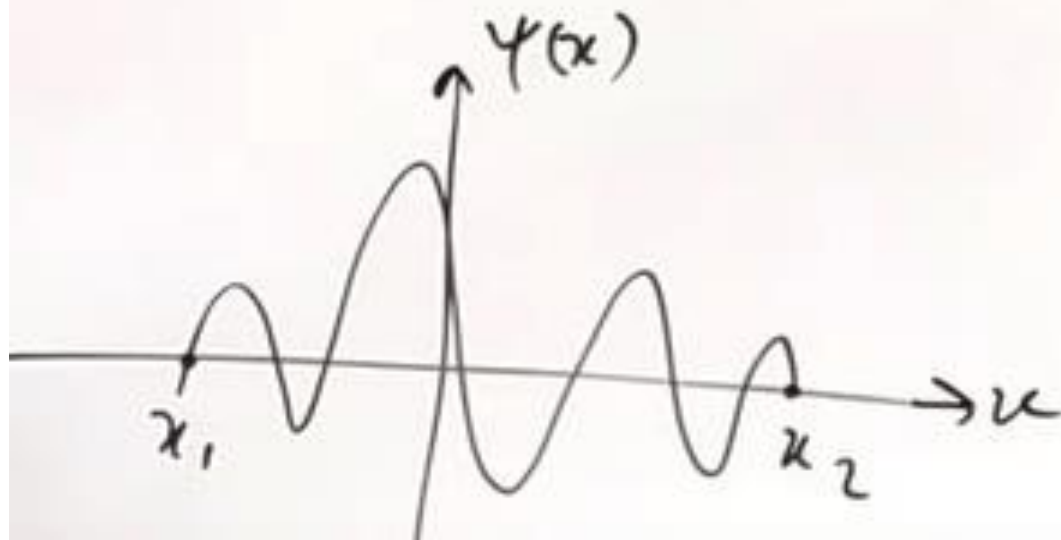


Unit-3

Lecture-3

Expectation values

Expectation value of any
Dynamical variable.



Since we know in Q.M. state of any particle is represented by normalized wave functⁿ associated with it. If we know state / wave functⁿ of any particle then we know everything about particle.

A wave function $\Psi(r, t)$ is said to be normalized if the probability of finding a quantum particle somewhere in a given space is unity

Everything $\rightarrow$ position of particle, momentum of particle, E , etc (Dynamical variable)

$\Rightarrow$ But due to probabilistic nature of wave functⁿ we can't determine exact value of Dynamical variable. so we find expectation value of Dynamical variable.

$\Rightarrow$ What do you mean by expectation value $\rightarrow$

• Expectation value is avg. of large no. of measurement in same system.

$$T_{avg} = \frac{T_1 + T_2 + \dots + T_{10}}{10}$$

$$[T_{avg} \approx T_{exact}]^{10}$$

$$T_p \approx T_T$$

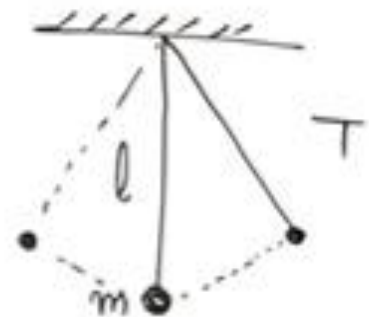
$$T_1 = 2 \text{ sec}$$

$$T_2 = 1.9$$

$$T_3 = 2.2$$

$$\vdots$$

$$T_{10} = 1.8$$



How to calculate Expectation value of any Dynamical variable:- # Expectation value of any

- Expectation value of any dynamical variable can be calculated Dynamical variable by following way.

let we want to find Expectation value any dynamical variable 'A'

$$\left[\langle A \rangle = \int \Psi^*(\vec{r}, t) \hat{A} \Psi(\vec{r}, t) d^3r \right]$$

$\Psi(\vec{r}, t) \rightarrow$ wave funⁿ of particle

$\Psi^*(\vec{r}, t) \rightarrow$ complex conjugate of $\Psi(\vec{r}, t)$

$\hat{A} \rightarrow$ operator corresponding to dynamical variable 'A'

Some dynamical variable & there operator.

D.V. position	op.
	$\hat{r}$ (3-D) $\hat{x}, \hat{y}, \hat{z}$ (1-D)
momentum	$\hat{p} = -i\hbar \nabla$ (3-D)
	$\left. \begin{aligned} \hat{p}_x &= -i\hbar \partial/\partial x \\ \hat{p}_y &= -i\hbar \partial/\partial y \\ \hat{p}_z &= -i\hbar \partial/\partial z \end{aligned} \right\}$ (1-D)
Energy	$\hat{E} = i\hbar \frac{\partial}{\partial t}$

Expectation value of position:- (r)

Defn- $\langle A \rangle = \int \Psi^*(\vec{r}, t) \hat{A} \Psi(\vec{r}, t) d^3r$

$\Psi(\vec{r}, t) \rightarrow$ must be normalized.

$$\langle \vec{r} \rangle = \int \Psi^*(\vec{r}, t) \hat{r} \Psi(\vec{r}, t) d^3r$$

In 1-D

$$\langle x \rangle = \int \Psi^*(x, t) \hat{x} \Psi(x, t) dx$$

$$\text{or } \langle x \rangle = \int \Psi^*(x, t) x \Psi(x, t) dx$$

$$\langle y \rangle = \int \Psi^*(y, t) \hat{y} \Psi(y, t) dy$$

$$\langle z \rangle = \int \Psi^*(z, t) \hat{z} \Psi(z, t) dz$$

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}, \quad \hat{p}_y = -i\hbar \frac{\partial}{\partial y}, \quad \hat{p}_z = -i\hbar \frac{\partial}{\partial z}$$

* Expectation value of any function $f(\vec{r}, t)$

$$\left[\langle f(\vec{r}, t) \rangle = \int \psi^*(\vec{r}, t) f(\vec{r}, t) \psi(\vec{r}, t) d^3r \right]$$

P.E. $V = V(\vec{r}, t)$

$$\left[\langle V \rangle = \int \psi^*(\vec{r}, t) V(\vec{r}, t) \psi(\vec{r}, t) d^3r \right] \text{ in 1-D.}$$

Expectation value of momentum:- (P) ...

$$\langle P \rangle = \int \psi^*(\vec{r}, t) \hat{p} \psi(\vec{r}, t) d^3r \quad \because \hat{p} = -i\hbar \nabla$$

$$\langle P \rangle = \int \psi^*(\vec{r}, t) (-i\hbar \nabla) \psi(\vec{r}, t) d^3r$$

$$\left[\langle P \rangle = -i\hbar \int \psi^*(\vec{r}, t) \nabla \psi(\vec{r}, t) d^3r \right] \rightarrow \text{in 3-D.}$$

$$\langle P_x \rangle = -i\hbar \int \psi^*(x, t) \frac{\partial}{\partial x} \psi(x, t) dx$$

$$\langle P_y \rangle = -i\hbar \int \psi^*(x, t) \frac{\partial}{\partial y} \psi(y, t) dy$$

$$\langle P_z \rangle = -i\hbar \int \psi^*(z, t) \frac{\partial}{\partial z} \psi(z, t) dz$$

H.W.

$$\langle E \rangle \quad \psi(\vec{r}, t)$$

Wave function of a free particle

A free particle is a particle on which no external force acts. (momentum, energy - constant)
potential independent of position)

$$F = -\nabla V$$

$$\leftarrow F_{\text{ext}} = 0$$

$$0 = -\frac{dV}{dx}$$

$$F = \frac{dp}{dt}$$

$$K.E = \frac{p^2}{2m}$$

$$0 = V \text{ is independent of } x$$

$$0 = p \text{ is constant}$$

Wave function of a free particle

A free particle is a particle on which no external force acts. (momentum, energy - constant)
potential independent of position)

$$\psi = A e^{\frac{i}{\hbar} (Et - px)}$$

$$(k) = \frac{2\pi}{\lambda} = \frac{2\pi p}{h} = \left(\frac{p}{\hbar} \right)$$

$$E = \hbar \omega$$

$$\omega = \frac{E}{\hbar}$$

$$\psi = A e^{i(kx - \omega t)}$$

$$\psi = A e^{i\left(\frac{p}{\hbar}x - \frac{E}{\hbar}t\right)}$$

$$\psi = A e^{\frac{i}{\hbar} (-px + Et)}$$

$$\psi = A e^{\frac{i}{\hbar} (Et - px)}$$